Nathaniel York

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EDUCATION

University of California, Davis — B.S. in Computer Science *Expected June 2027*

Relevant Coursework: Algorithm Design and Analysis, Data Structures, Database Systems, Probability & Statistical Modeling for CS, Object-Oriented Programming, Computer Architecture

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, Dart, SQL (basic), Bash

Frameworks & Tools: Flutter, Node.js/Express, Google App Scripts, Git, Figma, OpenAI API, Amazon SP-API, REST APIs

Concepts: Data Structures & Algorithms, OOP, API Integration, Data Processing & Automation, Database Fundamentals

WORK EXPERIENCE

Software Engineering Intern *Aug 2026 – Present*

Kaleon *Remote*

- Contribute across a 3-person engineering team building a transfer-planning platform helping community college students plan degree and transfer pathways
- Work on UX polish, accessibility readiness testing (keyboard navigation, screen-reader semantics, contrast), and demo reliability hardening ahead of institutional pilots
- Research and help architect a planned integration with ASSIST, California's official college articulation data source, including data mapping and a validation/fallback strategy

Software Engineer Intern *Sep 2024 – May 2025*

Design Hub at American River College *Carmichael, CA*

C++, Google App Scripts

- Developed and deployed a Discord bot automating server management and Google Drive organization for 200+ users
- Built data-scraping scripts integrating Google App Scripts to streamline documentation and team analytics
- Contributed to a cross-functional team improving workflow automation and version control across multiple projects

PROJECTS

NAB amp — Amazon FBA Sourcing & Sales Analytics Platform *Aug 2026 – Present*

Full-stack PWA | JavaScript, Node.js/Express, Amazon SP-API, Google Sheets API

- Designed and deployed a full-stack progressive web app integrating six Amazon Selling Partner API roles to automate live product-sourcing decisions — barcode/UPC scanning, listing-eligibility checks, and real-time profit/ROI calculations — replacing paid third-party tools
- Built a stateless sales-analytics sync pipeline pulling live order and settlement data via the Orders and Finance APIs, with refund-netting logic and a retry queue that resolved a production bug silently blocking order sync
- Deployed and in active daily use for real purchasing and sales decisions across 50+ transactions with a business partner; audited credential handling via git-history review

ML Trading Edge Discovery — Statistical Research Platform *2026 – Present*

Python, pandas, scikit-learn, LightGBM, SQLAlchemy, Streamlit

- Built a systematic research pipeline to statistically validate (not deploy) candidate trading strategies, with point-in-time feature computation designed to prevent look-ahead bias and purged walk-forward cross-validation with embargo periods
- Implemented Benjamini-Hochberg FDR correction and Deflated Sharpe Ratio as enforced statistical gates in a staged hypothesis-screening funnel across correlated instrument groups
- Maintained 258+ passing automated tests and reproducible experiment tracking (versioned migrations, config/git-hash snapshots) across an evolving research database

AI Image Renderer *Sep 2024 – Dec 2024*

Python, OpenAI API, Figma

- Engineered an AI-powered system to generate client design visuals using OpenAI's API
- Reduced concept generation time by 60% through prompt optimization and automation
- Delivered production-quality renders used in 3+ client campaigns at Design Hub

LEADERSHIP

Computer Science Club at American River College — Vice President *Sep 2024 – May 2025*

Carmichael, CA

- Directed one of ARC's largest clubs (50+ members) focused on AI, game development, and hackathons
- Hosted weekly LeetCode sessions improving members' problem-solving confidence and interview readiness
- Led semester-long project competitions resulting in 3 functional prototype builds

Coding for Kids at the Fair Oaks Library — Founder & Instructor *Aug 2024 – Dec 2024*

Fair Oaks, CA

- Designed and launched a free coding program for underprivileged students (grades 4–6)
- Taught Python and Scratch to 25+ children, increasing digital literacy and coding confidence
- Secured community space and resources to ensure zero-cost, full accessibility